COMMANDS FROM RTL TO GDS PRE SIGNOFF

1. create an env file include vcs, verdi, novas path
2. which vcs
3. which verdi
4. novas and Verdi has same path
5. export VCS_HOME=/path/to/vcs
6. export VERDI_HOME=/path/to/Verdi
7. export NOVAS_HOME=/path/to/novas
8. source env_file
9. vcs design.v tb.v -lca -kdb -debug_access+all -full64
10. ./simv
11. ./simv -verdi
12. cd DC/
13. dc_shell
14. source -echo -verbose ./rm_setup/dc_setup.tcl
15. set RTL_SOURCE_FILES ./../rtl/full_adder.v
16. define_design_lib WORK -path ./WORK
17. analyze -format verilog ${RTL_SOURCE_FILES}
18. open dc_shell usng command stat_gui
19. elaborate ${DESIGN_NAME}
20. stat_gui
21. Note open .sdc file
22. In terminal to know how mnaty clock type type get_porrts
23. To highlight your ports in design type change _selection [get_ports]
24. read_sdc ./../CONSTRAINTS/full_adder.sdc
25. note to know which lib working in tyoe get_lib
26. compile
27. compiler_ultra
28. note to know how many cells type type report_cells
29. note to know timing type type report_timing
30. write_sdc ./results/full_ader.sdc

31. you get .v gate levek netlist
32. note if you closed all files again opoen terminal ango to DC folder
33. dc_shell
34. set link_library {* ../ref/lib/stdcell_rvt/saed3rvt_tt0p78vnhoc.db }
35. set target_library { ../ref/lib/stdcell_rvt/saed3rvt_tt0p78vnhoc.db }
36. read_verilog ../rtl/full_adderr.v
37. start_gui
38. read_sdc ../CONSTRAINTS/full_adder.sdc

39. sorce ./run_dc.tcl
40. go to ICC2 folder
41. icc2_shell
42. set PDK_PATH ./../ref
43. create_lib -ref_lib $PDK_PATH/lib/ndm/saed32rvt_c.ndm FULL_ADDER_LIB
44. read_verilog {./../DC/results/full_adder.mapped.v} -library FULL_ADDER_LIB -design full_adder -top full_adder
45. stat_gui
46. click onn open layout when tool open
47. in terminal type initialize_floorplan
48. initialize_floorplan -core_offset 2.5
49. to save type save_lib
50. save_block
51. if you want to exit or open anytime type always open_lib FULL_ADDER_LIB1
52. open_block full_adder
53. start_gui
54. senarior 2
55. initialize_floorplan -core_offset 2
56. set_individual_pin_constraints -ports [get_ports] -sides 2
57. place_pins -self

58. create_placement -floorplan -effort very_low
59. we nee to create core power ring
60. type connect_pg_net -all_blocks -automatic
61. check vdd and vss created type to check get_ports if not type
62. create_net -power {VDD}
63. create_net -ground {VSS}
64. to check get_ports
65. create_pg_ring_pattern core_ring_pattern -horizontal_layer M7 -horizontal_width .4 -horizontal_spacing .3 -vertical_layer M8 -vertical_width .4 -vertical_spacing .3
66. set_pg_strategy core_power_ring -core -pattern {{name : core_ring_pattern}{nets : {VDD VSS}}{offset : {.5 .5}}}
67. compile_pg -strategies core_power_ring
68. create_pg_mesh_pattern mesh -layers { {{vertical_layer: M6}{width: .34} {spacing: interleaving}{pitch: 5} {offset: .5}} {{horizontal_layer: M7}{width: .38} {spacing: interleaving} {pitch: 5} {offset: .5}}
69. set_pg_strategy core_mesh -pattern { {pattern:mesh} {nets: VDD VSS}} -core -extension {stop: innermost_ring}
70. compile_pg -strategies core_mesh
71. create_pg_std_cell_conn_pattern std_cell_rail -layers {M1} -rail_width 0.09

72. set_pg_strategy rail_strat -core -pattern {{name: std_cell_rail} {nets: VDD VSS} }
73. compile_pg -strategies rail_strat
74. to check power type check_pg_drc
75. if check report type report_utlization
76. if yo want detail report type report_design -all
77. placement now
78. remove_modes -all; remove_corners -all; remove_scenarios -all
79. set mode1 "func"
80. set corner1 "nom"

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set scenario1 "${mode1}::${corner1}"
create_mode $mode1
create_corner $corner1
create_scenario -name func::nom -mode func -corner nom
to check if moe create or not type report_scenario
impotant  type source ./../CONSTRAINTS/full_adder.sdc
current_mode func
current_corner nom
current_scenario func::nom
set parasitic1 "p1"
set tluplus_file$parasitic1 "$PDK_PATH/tech/star_rcxt/saed32nm_1p9m_Cmax.tluplus"
set layer_map_file$parasitic1 "$PDK_PATH/tech/star_rcxt/saed32nm_tf_itf_tluplus.map"
set parasitic2 "p2"
set tluplus_file$parasitic2 "$PDK_PATH/tech/star_rcxt/saed32nm_1p9m_Cmin.tluplus"
set layer_map_file$parasitic2 "$PDK_PATH/tech/star_rcxt/saed32nm_tf_itf_tluplus.map"
read_parasitic_tech -tlup $tluplus_filep1 -layermap $layer_map_filep1 -name p1
read_parasitic_tech -tlup $tluplus_filep2 -layermap $layer_map_filep2 -name p2
set_parasitic_parameters -late_spec $parasitic1 -early_spec $parasitic2
set_app_options -name place.coarse.continue_on_missing_scandef -value true
type place_opt
now clock tree sysnthesis
synthesize_clock_tree
set_app_options -name cts.optimize.enable_local_skew -value true
set_app_options -name cts.compile.enable_local_skew -value true
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```
105.	set_app_options -name cts.compile.enable_global_route -value
    false
106.	set_app_options -name clock_opt.flow.enable_ccd -value true
107.	clock_opt
108.	set_app_options -name route.global.timing_driven -value true
109.	set_app_options -name route.global.crosstalk_driven -value
    false
110.	#track assignment
111.	set_app_options -name route.track.timing_driven -value true
112.	set_app_options -name route.track.crosstalk_driven -value true
113.	route_global
114.	#save_block route_global_database
115.	route_track
116.	#save_block route_track_database
117.	route_detail
118.	route_opt
119.	report_timing
120.	set fillers_ref "*/*SHFILL128_RVT */*SHFILL64_RVT
    */*SHFILL3_RVT */*SHFILL2_RVT */*SHFILL1_RVT"
121.	create_stdcell_fillers -lib_cell $fillers_ref
122.	writegs -merge_files {~
    /RTL2GS/ref/lib/gds/saed3nm_rvt_b.gs} full.gds
123.	or write_gds ./full.gds if no merge files
124.	read_gs ./full.gds
125.	open_block full_adderr
126.	start_gui
127.	PT
128.	Cd PT
129.	Invoke the tools by pt_shell
130.	set link_path "./../ref/lib/stdcell_rvt/saed32rvt_ss0p7vn40c.db"
131.	read_verilog "./../ICCII/results/full_adder.routed.v"
132.	link_design
133.	read_sdc ./../CONSTRAINTS/full_adder.sdc
134.	read_parasitics
    "./../ICCII/results/full_adder_func::nom.spef.p1_125.spef"
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```
update_timing -full
report_timing
report_design
check_timing
read_sdc ./../CONSTRAINTS/full_adder.sdc
update_timing -full
report_timing
report_design
check_timing
report_glo
start_gui
```